

# Hanson Smith

## Curriculum Vitae

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### Education

- 2022-Present **Assistant Professor**, California State University San Marcos  
2020-2022 **Assistant Research Professor**, University of Connecticut, Storrs  
2020 **Ph.D. Mathematics**, University of Colorado Boulder  
Thesis Title: *Monogeneity and Torsion*  
Advisor: Katherine E. Stange.  
2017 **M.A. Mathematics**, University of Colorado Boulder  
2014 **B.A. Mathematics**, Colorado College, Colorado Springs

### Research

#### Interests

Algebraic number theory and arithmetic geometry. Specifically, the monogeneity of algebraic number fields, i.e., the existence of a power integral basis for the ring of integers, torsion on arithmetic geometric objects, and the relationship between torsion and monogeneity.

#### Papers

- Preprint 2026 **Number Fields with Arbitrarily Large Minimal Index**  
arXiv: 2606.25841
- Preprint 2025 **Prime Splitting and Common  $N$ -Index Divisors in Radical Extensions: Part  $p = 2$**   
Written with Dylan Scofield.  
arXiv: 2512.23677
- Preprint 2025 **General Boyd–Lawton Theorems with Multivariable Limits**  
Written with Wayne Aitken and Kimberly Ayers.  
arXiv: 2508.05910
- 2026 **Critical Point Criteria and Dynamically Monogenic Polynomials**  
To appear in the Canadian Journal of Mathematics  
Written with Joachim König and Zack Wolske.  
DOI: 10.4153/S0008414X26102351
- 2026 **Prime Splitting and Common Index Divisors in Radical Extensions**  
To appear in Functiones et Approximatio Commentarii Mathematici  
DOI: 10.7169/facm/250821-9-2

- Preprint 2024 **Iterates of Quadratics and Monogenicity**  
 To appear in the Rocky Mountain Journal of Mathematics  
 Written with Zack Wolske.  
 arXiv: 2406.03629
- 2025 **Radical Dynamical Monogenicity**  
 Journal de Théorie des Nombres de Bordeaux, volume 37, number 1.  
 DOI: 10.5802/jtnb.1317
- 2023 **The Scheme of Monogenic Generators II: Local Monogenicity and Twists**  
 Research in Number Theory, volume 9, article 43.  
 Written with Sarah Arpin, Sebastian Bozlee, and Leo Herr.  
 DOI: 10.1007/s40993-023-00449-7
- 2023 **The Scheme of Monogenic Generators I: Representability**  
 Research in Number Theory, volume 9, article 14.  
 Written with Sarah Arpin, Sebastian Bozlee, and Leo Herr.  
 DOI: 10.1007/s40993-022-00419-5
- 2023 **Ramification in Division Fields and Sporadic Points on Modular Curves**  
 Research in Number Theory, volume 9, article 17.  
 DOI: 10.1007/s40993-023-00424-2
- 2022 **Frobenius Finds Non-monogenic Division Fields of Abelian Varieties**  
 International Journal of Number Theory, volume 18, number 10.  
 DOI: 10.1142/S1793042122501172
- 2022 **Monogenic Fields Arising from Trinomials**  
 Involve, volume 15, number 2.  
 Written with Ryan Ibarra, Henry Lembeck, Mohammad Ozaslan, and Katherine E. Stange.  
 DOI: 10.2140/involve.2022.15.299
- 2021 **Non-monogenic Division Fields of Elliptic Curves**  
 Journal of Number Theory, volume 228.  
[Video abstract.](#)  
 DOI: 10.1016/j.jnt.2021.03.024
- 2021 **A Divisor Formula and a Bound on the  $\mathbb{Q}$ -Gonality of the Modular Curve  $X_1(N)$**   
 Research in Number Theory, volume 7, article 22.  
 Written with Mark van Hoeij.  
 DOI: 10.1007/s40993-021-00243-3
- 2021 **The Monogeneity of Radical Extensions**  
 Acta Arithmetica, volume 198.  
 DOI: 10.4064/aa200811-7-10
- 2019 **Optimal Packings of Two to Four Equal Circles on Any Flat Torus**  
 Discrete Mathematics, volume 342.  
 Written with Madeline Brandt, William Dickinson, AnnaVictoria Ellsworth, and Jennifer Kenkel.  
 DOI: 10.1016/j.disc.2019.111597

- 2019 **A Family of Monogenic  $S_4$  Quartic Fields Arising from Elliptic Curves**  
 Journal of Number Theory, volume 197.  
 Written with T. Alden Gassert and Katherine E. Stange.  
 DOI: 10.1016/j.jnt.2018.09.026
- 2018 **Two Families of Monogenic  $S_4$  Quartic Number Fields**  
 Acta Arithmetica, volume 186.  
 DOI: 10.4064/aa180423-24-8

## Invited Talks

- *Monogenicity and Iteration*. USD Faculty Colloquium, University of San Diego, Dec. 9, 2025.
- *Critical Point Criteria for Dynamically Monogenic Polynomials*. Meeting on Monogeneity and Power Integral Bases, Universitat Politècnica de Catalunya, Barcelona, Spain, July 9, 2025.
- *Monogenic Number Fields by Example*. Simple Words Online Seminar, Dec. 4, 2024.
- *Monogenicity and Arithmetic Dynamics*. Leiden University Algebra, Geometry, and Number Theory Seminar, May 22, 2024.
- *Monogenicity: One Generator, Many Faces*. Front Range Number Theory Day, Apr. 13, 2024.
- *Richard Guy's Strong Law of Small Numbers and How Not to Make Friends*. HWY 78 Math Fields Day, Mar. 2, 2024.
- *On the Monogenicity of Iterated Polynomials*. Joint Mathematics Meetings, Jan. 4, 2024.
- *Radical Dynamical Monogenicity*. Online Conference on Monogeneity and Power Integral Bases, Sept. 7, 2023.
- *Monogenerators and Iteration*. Joint Mathematics Meetings, Jan. 6, 2023.
- *Radical Geometric Monogenicity*. UC Irvine Number Theory Seminar, Nov. 3, 2022.
- *Richard Guy's Strong Law of Small Numbers and How Not to Make Friends*. San Marcos Informal Mathematics In-person Colloquium, Sept. 8, 2022.
- *A Moduli Space of Monogenerators*. Online Number Theory Seminar, Jun. 3, 2022.
- *Richard Guy's Strong Law of Small Numbers and How Not to Make Friends*. Union College Student Seminar, Apr. 21, 2022.
- *Monogenicity and Torsion*. University of Rhode Island Math Colloquium, Mar. 4, 2022.
- *Trinomials, Monogenic Polynomials, and Questions*. Online Conference on Monogeneity and Power Integral Bases, Feb. 17, 2022.
- *Richard Guy's Strong Law of Small Numbers and How Not to Make Friends*. Western Colorado University Math, Computer Science, and Engineering Seminar, Nov. 29, 2021.
- *Monogeneity and Torsion*. University of Zagreb Seminar on Number Theory and Algebra, May 3, 2021.

- *Radical Monogeneity and Torsion*. UC Davis Algebra and Discrete Mathematics Seminar, Apr. 23, 2021.
- *Radical Monogeneity and Torsion*. Charles University Number Theory Seminar, Mar. 16, 2021. [Video](#).
- *Radical Monogeneity and Torsion*. Inaugural Online Conference on Monogeneity and Power Integral Bases, Jan. 14, 2021.
- *Ramification in Division Fields and Sporadic Points on Modular Curves*. AMS Special Session on Branching Out: Ramification Invariants in Algebra and Geometry. JMM, Jan. 8, 2021.
- *The Monogeneity of Division Fields of Abelian Varieties*. Uconn Algebra Seminar. Oct. 7, 2020.
- *Non-monogenic division fields of elliptic curves*. AMS Special Session on Algorithms, Experimentation, and Applications in Number Theory. JMM Denver, Jan. 16, 2020.
- *A Survey of Monogeneity*. Number Theory Seminar, CU Boulder, Dec. 3, 2019.
- *A Family of Monogenic  $S_4$  Quartic Fields Arising from Elliptic Curves*. Special session on algebraic number theory and Diophantine equations at the AMS Spring Central and Western Joint Sectional Meeting. University of Hawaii at Manoa, March 23, 2019.
- *Two Families of Monogenic  $S_4$  Quartic Number Fields*. Hawaiian Number Theory (HINT). University of Hawaii at Manoa, March 20, 2019.
- *Ramification in the Division Fields of Supersingular Elliptic Curves and Sporadic Points on Modular Curves*. Number Theory Seminar, CU Boulder, September 4, 2018.
- *A Historical Motivation for Quadratic Reciprocity*. Fearless Friday, Colorado College, March 3, 2017.

## Other Research Talks and Presentations

- *A Research Story: Critical Points and Dynamical Monogenicity*. San Marcos Informal Mathematics In-person Colloquium, CSUSM, Sept. 25, 2025.
- *Critical Point Criteria and Dynamical Monogenicity*. Journées Arithmétiques, University of Luxembourg, Luxembourg, July 3, 2025.
- *Critical Points and Dynamically Monogenic Polynomials*. Special Session on Number theory at Primarily Undergraduate Institutions, AMS Sectional Meeting, Cal Poly San Luis Obispo, May 3, 2025.
- *Radical Dynamical Monogenicity*. CSU Mathematical Sciences Conference, CSU Bakersfield, Nov. 11, 2023.
- *Power Bases and Moduli Spaces: A Scheme of Monogenerators*. CSU Joint Mathematics Meetings, CSU Northridge, Nov. 11, 2022.
- *Frobenius and the Monogeneity of Division Fields of Abelian Varieties*. Upstate Number Theory Conference, Union College, Oct. 23, 2021.
- *Non-Monogenic Division Fields and Endomorphisms of Abelian Varieties*. Young Researchers in Algebraic Number Theory, Zoom, August 18, 2021.

- *Richard Guy's Strong Law of Small Numbers and How Not to Make Friends*. UConn Math Club talk, Zoom, September 23, 2020.
- *Non-monogenic Division Fields of Elliptic Curves*. Junior Mathematician Research Archive recorded talk, posted to YouTube September 1, 2020. Link: <https://www.youtube.com/watch?v=2wkRLvfkfrs>.
- *An Upper Bound on the  $\mathbb{Q}$ -gonality of  $X_1(N)$* . Lightning talk at Front Range Number Theory Day, Zoom, April 25, 2020. I also was an organizer of this conference.
- *The Monogeneity of Kummer Extensions and Radical Extensions*. West Coast Number Theory, Pacific Grove, CA, Dec. 17, 2019. I also chaired a session at this conference.
- *Problems with Number Fields*. Graduate Student Seminar, CU Boulder, Dec. 4, 2019.
- *The Monogeneity of Kummer Extensions and Radical Extensions*. Number Theory Series in Los Angeles. Occidental College, Oct. 26, 2019.
- *Non-monogenic Division Fields of Ordinary Elliptic Curves*. Lightning talk for the Arithmetic of Low-Dimensional Abelian Varieties workshop at ICERM of Brown University, June 4, 2019.
- *Ramification in the Division Fields of Supersingular Elliptic Curves and Sporadic Points on Modular Curves*. Canadian Number Theory Association Conference, Université Laval, July 12, 2018.
- *Ramification in the Division Fields of Supersingular Elliptic Curves and Sporadic Points on Modular Curves*. Connecticut Summer School in Number Theory, The University of Connecticut, June 2, 2018.
- *Monogenic  $S_4$  Quartic Fields Arising from Elliptic Curves*. Strength in Numbers, Queen's University, May 11, 2018.
- *Transcending the Irrationality of Pi: Hilbert's Seventh Problem*. Slow Pitch, CU Boulder, March 14, 2018.
- I presented a poster on *A family of monogenic  $S_4$  quartic fields arising from elliptic curves*. Southern California Number Theory Day, The University of California Irvine, Oct. 21, 2017.
- *The Group Law on Elliptic Curves using Riemann-Roch*. Slow Pitch, CU Boulder, Nov. 16 and 23, 2016.
- *A Proof of Quadratic Reciprocity*. Slow Pitch, a graduate student colloquium at CU Boulder, April 20, 2016.
- *Fermat's Last Theorem, Elliptic Curves, Circle Packings, and Modular Curves*. Thesis Presentation, Colorado College, May, 2014.
- *Packings of Four Equal Circles on Flat Tori*. Mathfest Undergraduate Paper Session, with Madeline Brandt, August, 2013.

## Workshops and Professional Development

- Pens, Processors, and GenAI: Writing Has Always Been Tool-Mediated. Mini-FLC run by Dr. Mary Stewart. Spring 2026.
- K-16 Math for Careers Summit. Organizing Committee member, facilitator, and panelist. Sept. 12, 2024.

- Park City Mathematics Institute Research Program. Summer 2022.
- Inclusive STEM Teaching Learning Community. Online, spring 2022.
- Virtual Inquiry Based Learning Workshop run by the Academy of Inquiry Based Learning. June 23-26, 2020.
- TRansforming Instruction in Undergraduate Mathematics via Primary Historical Sources (TRIUMPHS) Workshop. New Mexico State University, July 19-20, 2019.
- Arizona Winter School on Topology and Arithmetic. The University of Arizona, March 2-6, 2019.
- Arizona Winter School on Iwasawa Theory. The University of Arizona, March 3-7, 2018.
- Pacific Institute for the Mathematical Sciences (PIMS) Workshop on Computational Arithmetic Geometry. Simon Fraser University, June 5-9, 2017.

## ■ Outreach Talks and Panels

- *Euclid's Elements and Sidewalk Chalk Constructions*, CSTEM Summer Camp for high school students. June 24, 2026.
- *Math Research and Preparing for Math Success*, Early Assessment Program High School Counselor Breakfast. Mar. 10, 2023.
- *Richard Guy's Strong Law Of Small Numbers And How Not To Make Friends*, UConn Math Club. Sept. 23, 2020.
- *Colorado College SIAM Virtual Graduate School Information Panel*, panelist. Colorado College, Nov. 5, 2020.
- *The Difficulty of Putting Sprinkles on Donuts*. 3 Minute Thesis Competition Finals, CU Boulder, Jan. 31, 2020.
- *The Difficulty of Putting Sprinkles on Donuts*. 3 Minute Thesis Competition Preliminary Round, CU Boulder, Nov. 8, 2019.
- *The Difficulty of Putting Sprinkles on Donuts*. Lightning talk at the Research and Innovation Week. CU Boulder, Oct. 16, 2019. (This talk won the Research and Innovation Week lightning talks session.)
- *Number Theory, Geometry, Elliptic Curves, and a Piece of the Pie*. STEMinar: A seminar for coached general audience talks, CU Boulder, February 21, 2019.
- *Putting Sprinkles on Donuts*. 3 Minute Thesis Competition, CU Boulder, fall 2018.
- *Number Theory, Geometry, Elliptic Curves, and a Piece of the Pie*. Colorado Academy Math Club Invited Talk. Denver, CO, October 30, 2018.
- *Number Theory, Geometry, Elliptic Curves, and a Piece of the Pie*. Great Talks for a General Audience: Coached Presentations for Graduate Students. MathFest Denver, CO, August 4, 2018.
- *Beyond an Undergraduate Mathematics Degree*. I was a panelist in the aforementioned panel at the Pikes Peak Regional Undergraduate Mathematics Conference, University of Colorado Colorado Springs, Feb. 24, 2018.
- *Number Theory: A Problematic Introduction*. Initial Conditions, a seminar for first-year graduate students at CU Boulder, Nov. 3, 2017.

- *Richard Guy's Strong Law of Small Numbers, Numerology, and How Not to Make Friends*. Inaugural Tangents Colloquium, CU Boulder, April 14, 2017.

## Teaching and Coordinating

- MATH 100: Mathematical Ideas, CSUSM, fall 2025. I am responsible for a class of 39 students, and I am coordinating four additional sections.
- MATH 470: Abstract Algebra, CSUSM, fall 2025. I am responsible for a class of 15 students.
- MATH 520: Graduate Algebra, CSUSM, fall 2025. I am responsible for a class of around 14 students.
- MATH 260: Calculus III, CSUSM, spring 2025. I was responsible for two classes of around 30 students.
- MATH 100: Mathematical Ideas, CSUSM, spring 2025. I coordinated two sections.
- MATH 100: Mathematical Ideas, CSUSM, fall 2024. I was responsible for a class of 38 students, and I coordinated four additional sections.
- MATH 110: Critical Thinking, CSUSM, fall 2024. I was responsible for a class of 21 FTF students.
- MATH 470: Abstract Algebra, CSUSM, fall 2024. I was responsible for a class of 19 students.
- MATH 100: Mathematical Ideas, CSUSM, spring 2024. I coordinated four sections with a total of 151 students.
- MATH 378: Number Systems, CSUSM, spring 2024. I was responsible for a class of 24 students.
- MATH 470: Abstract Algebra, CSUSM, spring 2024. I was responsible for a class of 29 students.
- MATH 522: Number Theory, CSUSM, spring 2024. I was responsible for a class of 18 students.
- MATH 264: Introduction to Linear Algebra, CSUSM, fall 2023. I coordinated three sections with a total of 84 students.
- MATH 110: Critical Thinking, CSUSM, fall 2023. I was responsible for a class of 18 FTF students.
- MATH 378: Number Systems, CSUSM, fall 2023. I was responsible for a class of 26 students.
- MATH 378: Number Systems, CSUSM, spring 2023. I was responsible for a class of 30 students.
- MATH 470: Abstract Algebra, CSUSM, spring 2023. I was responsible for a class of 31 students.
- MATH 110: Critical Thinking, CSUSM, fall 2022. I was responsible for a class of 15 FTF students.
- MATH 470: Abstract Algebra, CSUSM, fall 2022. I was responsible for a class of 39 students.
- Algebraic Number Theory, Connecticut Summer School in Number Theory, UConn, summer 2022. I taught a four lecture mini-course giving a rapid introduction to algebraic number theory.
- Introduction to Number Theory, UConn, spring 2022. I was responsible for a class of 25 students.

- Honors Linear Algebra, UConn, spring 2022. I was responsible for a class of 35 students.
- Independent Study in Algebraic Number Theory, UConn, spring 2022. I advised two students with the goal gaining some perspective on algebraic number theory.
- Introduction to Number Theory, UConn, fall 2021. I was responsible for a class of 25 students.
- Linear Algebra, UConn, fall 2021. I was responsible for a class of 40 students.
- Linear Algebra, UConn, spring 2021. I was responsible for two classes of 35 students.
- Linear Algebra, UConn, fall 2020. I was responsible for two classes of 35 students.
- Calculus 3, CU Boulder, spring 2020. I was responsible for a class of 32 students.
- Calculus 3, CU Boulder, fall 2019. I was responsible for a class of 32 students.
- Abstract Algebra 1, Colorado College, spring 2018. I was invited back to my alma mater to co-teach this course with Beth Malmskog.
- Calculus 2, CU Boulder, fall 2017. I was responsible for a class of 32 students.
- Calculus 2, CU Boulder, spring 2017. I was responsible for a class of 32 students.
- Precalculus, CU Boulder, summer 2016. I was responsible for all aspects of the first half of this 32 student course.
- Calculus 1, CU Boulder, spring 2016. I was responsible for a class of 32 students.
- Calculus 1, CU Boulder, fall 2015. I was responsible for a class of 32 students.
- 2017-spring 2020: Mentorship of a local, mathematically-advanced high school student. This involved teaching my mentee algebraic and analytic number theory once a week as well as advising them on academic decisions.

## Research Directed

- I am currently advising a master's student, Dylan Scofield. He has advanced to candidacy and will defend his thesis in spring 2027.
- In the summer of 2024, I directed a four undergraduates and one graduate mentor in the Summer Scholars program in research on integral bases for cubic number fields. Students presented a poster and gave two talks.
- In the summer of 2018, my advisor and I ran an REU with three University of Colorado students studying the monogeneity of trinomials. The research program was based on my previous work and led to new results. See above for the resulting paper.

## Service and Leadership

- Co-Chair of the Math Department Hiring Committee 25-26. CSUSM, 2025-2026 academic year.
- CSTEM representative to the Faculty Awards Selection Committee. 2025-2026 academic year.
- CSUSM representative to the North Country Higher Education Alliance (NCHEA). 2025-present.
- Member of the Reid Lecture Committee. CSUSM, 2023-present.

- Co-Organizer of the San Marcos Informal Mathematics In-person Colloquium (SMIMIC). CSUSM, 2023-present.
- Mathematics Undergraduate Faculty Advisor, CSUSM, spring 2023-present. I am responsible for helping advise undergraduate math majors, especially those in the education option, on their academic and career plans.
- Reviewer for *Research in Number Theory*, *Journal de Théorie des Nombres de Bordeaux*, *Acta Arithmetica*, *Communications in Algebra*, *The New York Journal of Mathematics*, *The Rose-Hulman Undergraduate Mathematics Journal*, *The Asian-European Journal of Mathematics*, *Publicationes Mathematicae Debrecen*, *European Journal of Mathematics*, *Hacettepe Journal of Mathematics and Statistics*, *Commentationes Mathematicae Universitatis Carolinae*, *The Polynesian Journal of Mathematics*, and *The Journal of the Australian Mathematical Society*.
- Co-Chair of the Math Department Hiring Committee 23-24. CSUSM, 2023-2024 academic year.
- Joint Mathematics Meetings Undergraduate Student Poster Session judge. Denver, Jan. 17, 2020; Boston, Jan. 7, 2023; San Francisco, Jan. 5, 2024.
- Math Department Representative for the Dean's Committee on Outreach and Recruitment. CSUSM, spring 2023.
- **Super STEM Saturday**. Organizer of the the math department booth. CSUSM, Mar. 11, 2023.
- Co-founder and organizer of the **Front Range Number Theory Day**, a day long gathering of number theorists of all levels from around the Colorado Front Range. I organized and co-wrote the Research and Innovation Office (REO) grant that funded the spring 2019 number theory day at CU. I also co-wrote our successful NSF grant; see below.
- Organizer of a graduate student and faculty seminar on Bjorn Poonen's book "Rational Points on Varieties." CU Boulder and Colorado State University, spring 2020.
- Organizer of a graduate student and faculty seminar on the proof of Fermat's Last Theorem. CU Boulder, fall 2018.
- Cofounder and organizer of the Tangents Colloquium at CU. This more informal colloquium intended to highlight interesting, but not necessarily research level, topics. A large aim of this colloquium was to encourage conversations between faculty and graduate students.
- Representative for CU Boulder, JMM Graduate Fair, JMM 2017.

## Grants

- Site Tester for a new primary source project, part of the **TRIUMPHS** project on undergraduate math instruction via primary sources. Fall 2020.
- Co-Principal Investigator on an NSF grant for Conferences and Workshops in the Mathematical Sciences: The grant provided three years of funding for the Front Range Number Theory Day.

## Awards and Honors

- CSTEM Faculty Outreach Award Recipient. CSUSM, 2024-2025 academic year.
- CSTEM Mathematics Outstanding Faculty-Student Collaboration Award with Dylan Scofield. CSUSM, 2024-2025 academic year.
- CSTEM Faculty Outreach Award Nominee. CSUSM, 2023-2024 academic year.
- Graduate School Summer Doctoral Research Fellowship. CU Boulder, 2019.
- Magna Cum Laude with Distinction in Mathematics. Colorado College, 2014.
- Sophie Germain Prize in Mathematics. Colorado College, 2014.

## Research as an Undergraduate

- In the summer of 2013, I worked with another undergraduate at the Grand Valley State University mathematics REU under the guidance of Professor William Dickinson. Our work contributed to *Optimal Packings of Two to Four Equal Circles on Any Flat Torus*; see above.
- I was awarded a Collaborative Faculty-Student Research Venture Grant with Colorado College professor Stefan Erickson for work during the summer of 2012.